# MILITARY SPECIFICATION

# TUBES, RUBBER, PENROSE, DRAINAGE

## 1. SCOPE

1.1 Scope. This specification covers rubber tubes, of the Penrose design, suitable for drainage during or subsequent to surgical procedures.

1.2 Classification. Tubes shall be of the following sizes (nominal inside diameters), as specified (see 6.1):

Size 1 - 3/8 inch
Size 2 - 5/8 inch
Size 3 - 7/8 inch

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-566 | Boxes, Folding, Paperboard. |
| PPP-B-585 | Boxes, Wood, Wirebound. |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | Box, Fiberboard. |
| PPP-B-676 | Boxes, Set-Up, Paperboard. |

Military

| | |
|---|---|
| MIL-L-10547 | Liners, Case, Waterproof. |

STANDARDS

Federal

Fed. Test Method Std. 601 Rubber: Sampling and Testing.



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Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The drainage tubes shall be latex rubber.

3.2 Physical properties. The minimum physical properties of the rubber drainage tube before aging shall be as follows:

Tensile strength . . . . . 3500 pounds per square inch
Ultimate elongation . . . . . 800 percent
Tensile and elongation tests shall be performed as specified in 4.4.2.

3.3 Resistance to aging. The tensile strength and ultimate elongation shall not decrease more than 25 percent of the value before aging nor shall the rubber latex become tacky or brittle. Accelerated aging test shall be performed as specified in 4.4.3.

3.4 Style, design, and dimensions. The tube shall be 36 ± 1 inches long and of nominal inside diameter as indicated for the appropriate size (see 1.2). Tube shall have a wall thickness of 0.011 ± 0.003 inch and shall be capable of accepting a tape insert (not supplied) to serve as a drain. The flat width of the tubes shall be as follows:

| Size | Flat width (± 1/32 inches) |
|---|---|
| 1 | 9/16 |
| 2 | 15/16 |
| 3 | 1-9/32 |

Dimensions shall be measured as specified in 4.5.

3.5 Identification marking. Each tube shall be legibly and permanently marked with the size, contractor's name or registered trade-mark, and with date of manufacture (month-year).

3.6 Workmanship. Workmanship shall be first class throughout. Tubes shall be free from defects which detract from their appearance or may impair their serviceability.



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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 ampling and inspection criteria.

4.2. Inspection lot. Formation, size, and presentation of each inspection lot shal be in accordance with the requirements set forth in section 5 of Standard MIL-STD-105.

4.2.2 Sampling and lot disposition criteria. Sampling and disposition of each lot shall be determined in accordance with the procedures set forth in Standard MIL-STD-105.

4.2.2.1 For visual examination. Sampling and disposition of visual characteristics shall be determined from the following Inspection and Acceptable Quality Levels (AQL's):

| Category of Defect | Inspection Level | AQL (percent defective) |
|---|---|---|
| Major | II | 1.0 |
| Minor | II | 2.5 |

4.2.2.2 For test, dimensional, and preparation for delivery requirements. Sampling and disposition of test, dimensional, and preparation for delivery characteristics shall be determined from the following:



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| Component | Characteristic | Requirement | Procedure | Inspection level | AQL (percent defective) |
|---|---|---|---|---|---|
| 501 - Tube | Physical properties before aging | 3.2 | 4.4.1 and 4.4.2 * | L1 | 6.5 |
| 502 - Tube | Deterioration and physical properties after aging | 3.3 | 4.4.1 and 4.4.3 * | L1 | 6.5 |
| 503 - Tube | Dimensions | 3.4 | 4.5 * | I2 | 1.5 |
| 504 - Exterior shipping container and contents | Packaging<br>Packing<br>Marking<br>Levels of protection | ** | | L1 | 6.5 |

* Tubes expended as a result of wall thickness measurement may be used for physical testing provided that sufficient material remains to prepare required number of specimens.

** See contract and applicable paragraphs of section 5.

4.3 Inspection requirements. The tubes shall be examined to determine compliance with all requirements contained in this specification. The extent of testing, measurement, and examination listed in 4.2.2.2 above, and possible defects listed in table I, below, indicate minimum inspection requirements. Inspection is, however, not restricted to the defects listed in table I.

Table I. Classification of Defects

| Categories | Defects |
|---|---|
| **Major** | |
| 101 | Tube not free from holes, cuts, tears, slits, or perforations. |
| 102 | Surface of tubing not smooth and uniform; not free from internal or external tackiness, brittleness, open blisters, lumps, and embedded or ingrained foreign material. |
| 103 | Internal passage not clear and free from obstruction. |
| 104 | Identification marking not correct, complete, legible, and permanent. |
| **Minor** | |
| 201 | Tube not free from surface dirt, stains, or discoloration. |
| 202 | Tube not free from closed blisters. |



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4.4 Test procedures.

4.4.1 Sample and specimen preparation. Each sample tube selected in accordance with 4.2.2.2, for determining physical properties and resistance to accelerated aging, shall be cut into halves and divided into two separate groups for testing both before and after aging. The number of specimens referenced in 5.2 of method 4111 in Federal Test Method Standard No. 601 shall be prepared from each tube half in each group. Dumbbell specimens shall be utilized.

4.4.2 Physical properties. Tensile strength and ultimate elongation tests shall be performed in accordance with methods 4111 and 4121, respectively, of Federal Test Method Standard No. 601.

4.4.3 Resistance to aging. The accelerated aging test shall be performed in accordance with method 7221 of Federal Test Method Standard No. 601.

4.5 Measurement procedures.

4.5.1 Length. The tube shall be placed on a flat surface so as to expose its full length. A tape or scale graduated in 1/16 inches shall be applied parallel to the axis of the tube. Two readings shall be taken, one at each extremity of the width. Tubes on which both length measurements are not within tolerance as specified in 3.4 shall be considered defective.

4.5.2 Wall thickness. Wall thickness shall be measured by utilizing method 2011 of Federal Test Method Standard No. 601, except that two readings shall be taken at each end of the tube (four in all per unit) in a nondestructive manner where possible. Tubes on which the average of the four readings are not within the tolerance as specified in 3.4 shall be considered defective.

4.5.3 Flat width. The tube shall be placed on a flat surface. A scale graduated in 1/32 inches shall be applied at right angles to the axis of the tube. Sufficient pressure shall be exerted on the scale to assure that internal surfaces are in contact with each other across the full width of the tube. Four random readings shall be taken along the length of the tube. Tubes on which the average of the four width readings are not within tolerance as specified in 3.4 shall be considered defective.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) length of tubing, of the stock number specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table II shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.



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Table II. Packaging Quantities

| Unit package | Intermediate package | Exterior container |
|---|---|---|
| 1 unit | 12 units | 288 units |

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be folded not more than once over its length in such manner that it will not kink or be permanently deformed, and shall be packaged in a bag of appropriate size fabricated of glassine paper, polyethylene, cellophane, or other plastic film of suitable weight. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.3.2 Intermediate package. Twelve (12) unit packages (bags) shall be packaged flat (not folded) in a carton of appropriate size constructed in accordance with Specification PPP-B-566 or PPP-B-676. As an alternate, each of the twelve (12) unit packages (bags) may be folded not more than once over its length in such a manner that packaged unit will not kink, be permanently deformed, or unit bag damaged (scored or cut) and fitted into a carton of appropriate size constructed in accordance with Specification PPP-B-566 or PPP-B-676. Carton shall be adequately secured to prevent accidental opening.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load and constructed in accordance with Specification PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; or PPP-B-621, table III.

5.2.1.2 Case liner. Each level A box shall be lined with a waterproof case liner conforming to Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 Strapping. Strapping, when required, shall be in accordance with applicable box specification.

NOTE: Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.



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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be constructed in accordance with Specification PPP-B-621, table I, or PPP-B-636, type 1, class 1, fabricated from not less than 275-pound test fiberboard.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked with stock number, item identification, unit, quantity, and name of contractor.

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with 5.2 of Standard MIL-STD-129. Contract number and name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with Standard MIL-STD-129, except as follows: The letter "L" shall precede the Federal stock number. One space shall be allowed between the letter "L" and the first digit of the stock number. The terms "Stock number" or "FSN" shall not be indicated. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of the specification.
(b) Size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

6.3 Transportation description. Transportation description applicable to this item is:

Rubber tubing, NOIBN, in boxes.

Carload minimum weight 30,000 pounds per Uniform Freight Classification.

Truckload minimum weight 30,000 pounds per National Motor Freight Classification (A Series).



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